

10.7.21 Matgeo

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Question

Find the points on the curve $x^2 + y^2 - 2x - 3 = 0$ at which the tangents are parallel to the x-axis.

For the given curve

$$\mathbf{u} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \quad (1)$$

$$f = -3 \quad (2)$$

Solution

For a circle the points of contact are given by the formula :

$$\mathbf{q}_{ij} = \left(\pm r \frac{\mathbf{n}_j}{\|\mathbf{n}_j\|} - \mathbf{u} \right) \quad (3)$$

Where $i, j = 1, 2$

Since tangents are parallel to x - axis ,the normal is given as

$$\mathbf{n} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad (4)$$

Solution

Solving 0.5 we get :

$$\mathbf{q} = (\pm 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} -1 \\ 0 \end{bmatrix}) \quad (5)$$

$$= \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad (6)$$

Hence the two points of contact are

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad (7)$$

Graphical Representation

